Predicting Conversions using Machine Learning

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***Abstract*— Over the last few years, conversion prediction has emerged as a high-profile problem in digital advertising and marketing analytics but traditional predictive pipelines tend to maximize attributed conversions as opposed to actual incremental impact. The review is an overview of the recent machine learning methods to convert prediction and allocation of ads, focusing on the transition of supervised prediction models to causal uplift modeling and contextual decision policies. This paper will initially review popular supervised techniques such as logistic regression, gradient-boosted trees, and deep learning architectures and the strengths related to both large-scale tabular and behavioral data and limitations in case of class imbalance, delayed feedback, and calibration drift. It then discusses causal and uplift modeling frameworks as methods of estimating incremental conversion probability, considering the treatment-control design, propensity correction, and doubly robust estimation as a better method of optimization. The review also examines contextual multi-armed bandit-based approaches to adaptive budget and impression allocation which would be particularly useful in a non-stationary environment with delayed conversions and operational limits like CPA and pacing targets. Besides that, the evaluation practices are contrasted in offline metrics of prediction, uplift-specific analogy, and online incrementality. Lastly, the paper presents emerging issues discussing privacy-sensitive measurement, data sparsity, and deployment strength and research directions of incorporating causal inference, online learning, and limited optimization in real conversion systems. This review contributes to the systematic formulation of the methodology that can be adopted by researchers and practitioners interested in developing credible incrementality-sensitive, and privacy-conscious conversion optimization pipes.**

Keywords*— *Conversion Prediction; Digital Advertising; Machine Learning; Uplift Modeling; Incrementality; Contextual Multi-Armed Bandits; iROAS; Delayed Conversions, Privacy-Preserving Measurement.

I. Introduction

One of the most fundamental issues in ad-tech and mar-tech is conversion prediction since the process of media buying, audience targeting, and campaign optimization all require estimates of the probability of an impression or a click resulting in a valuable downstream event (purchase, sign-up, install, etc.). The conversion rate prediction has also been the focus of recent literature as the core of the effective advertising decision system particularly since models can now also run on a large scale when using sparse behavioral cues and evolving user paths [1]. Practically, conversion prediction does not just affect bidding and ranking, but also budget allocation, pacing and creative rotation, making business reliability of the model business-critical, and not just an academic goal [2], [3].

Conventional attribution-based optimization, however, does have critical constraints. Most of the production systems are conditioned to forecast attributed conversions (such as post-click or post-view labels) as opposed to the incremental conversions brought about by the very exposure to the ad. This leads to a discrepancy between the measured and the cause that marketers are interested in: causal lift. Even attribution is subject to conversion, which means that an attribution-based model can overvalue users who were already likely to convert. This is one of the reasons why uplift modeling and causal inference methods have gained relevance in marketing decision pipes [4]. Practical concerns, including sample-selection bias and data sparsity, are also mentioned in other related CVR literature, when training models only on click-space data but testing on the full impression space [3].

When optimization is done directly based on model outputs it is evident that the non-incremental conversion problem is more critical. The fact that the conversion probability is high does not always mean that its incremental effect of presenting an ad is high. That is to say, the quality of prediction is not enough in the case the decision objective is lift, iROAS, or efficient allocation of spend. The importance of causal and uplift modeling models lies in their explicit estimation of the heterogeneity in the treatment effect as opposed to the observed one when exposed [4]. This difference in performance marketing environments will prevent wasting money on certain users who are sure converters and will aid in more effective targeting of persuadable users.

Meanwhile, contemporary conversion optimization cannot be done with simple predictions in place or offline causal scoring. The advertising environment is volatile: auctions change, creators become tired, the behavior of the audience changes, and the feedback on conversions is usually delayed. Contextual bandit systems deal with it by jointly predicting and adopting exploration-exploitation policies, and offline assessment methods such as replay and doubly robust estimation assist in evaluating candidate policies prior to risky digital implementation [5], [6]. Another source of difficulty is delayed feedback since conversions can be received hours or days following impression or click events and this can be biased to learning unless mitigated; delayed-conversion bandit models and delayed-feedback contextual bandit analysis directly face this operational fact [7].

Another problem is measurement that is limited by privacy. Changes in platforms and browsers have minimized the supply of cross-site and user level identifier, which has rendered traditional attribution and user level tracking to be more difficult to sustain. The Apple App Tracking Transparency (ATT) system features a mandatory user permission to be tracked in the apps and sites of other companies, which has a significant impact on measuring mobile advertisements [8]. Parallel to this Google Privacy Sandbox Attribution Reporting API is designed to enable conversion measurement by means of privacy-compliant and browser-generated reports that do not share sensitive information [9]. These adaptations do not eliminate the necessity of conversion prediction; they just create additional demands of the powerful, privacy-sensitive, and causally worthy methods of measurement and optimization.

The given review has been helpful, as it links 4 strands which are frequently debated individually: (i) supervised conversion prediction models, (ii) causal/uplift estimation of incrementality, (iii) contextual bandits of adaptive decision-making, and (iv) privacy-aware measurement constraints. It offers an orderly perspective of the comparison of techniques not only in the degree of predictive power, but also in whether they assist in the quality of practical decisions with delayed results, resource pressures, and signal distortion [1], [4]–[7], [9].

Research gap and Questions

Even though considerable advancement has been made regarding predicting conversions, there are still many studies that will optimize on attributed conversions instead of actual added value. This can lead to businesses making poor decisions on ad spend. Additionally, the majority of current work treats the topics of prediction and causal uplift estimates separately from creating budget-constrained decision rules for both. These are rarely integrated into a single framework for real-world applications. Lastly, current research has failed to address issues concerning privacy-related losses of signal and delayed conversion along with creating a gap where no frameworks exist for developing incrementality-aware optimizations at a level that is sufficiently robust to be deployed.

RQ1: What are the most commonly used machine learning models in conversion prediction in digital advertising systems?

RQ2: To what extent do causal and uplift model techniques outperform incremental decision quality over attribution-based prediction?

RQ3: What are the ways of integrating budget, pacing, and CPA constraints into adaptive decision systems like the contextual bandits?

RQ4: Which are the significant research and implementation gaps of conversion optimization in the presence of privacy-limited measurement conditions?

II. REVIEW OF LITERATURE

A. Supervised Conversion Prediction and Feature-Interaction Modeling

Early industrial conversion modeling used both generalized linear models and engineered cross-features extensively, but the scale and sparsity of advertising logs drove the field to the representation learning and automated interaction discovery. One of the significant directions of research is concerned with learning feature interactions that model useradcontext interactions without their complete manual crossing. DeepFM is a unified factorization-machine style low-order interaction with deep neural higher-order interaction in a single end-to-end model and extensively used as a practical baseline in sparse categorical click/conversion models [10]. Related architectures focus on explicit crossing: the Deep and Cross Network (DCN) proposes a cross network that implements bounded-degree interaction in a way that is more accurate and that model complexity can be controlled in production-scale CTR/CVR prediction pipelines [11].

A second stream is marked by behavior-conscious modeling, which is driven by the fact that the probability of conversion is conditioned by the previous actions of users and their relation to the ad being displayed. Deep Interest Network (DIN) provides a solution to this problem by learning an adaptive representation of user interest through attention-like activation of the behaviors on the historical candidate ad in order to enhance expressiveness in large-scale e-commerce advertising settings [12]. In each of these models, the literature has been consistent in pointing out that the benefits of performance come not only due to deeper networks, but due to structural biases, which encode interaction patterns and behavior relevance, which is important when labels are rare, and features are in high-dimensionality.

In conversion-specific environments, mismatch between training distributions and serving distributions has also been studied by researchers. Traditional CVR models are usually trained on clicked samples, but are deployed throughout the entire impression space, which results in sample-selection bias and severely sparse conversion labelling. The Entire Space Multi-Task Model (ESMM) is a model that redefines the estimation of CVR by jointly modeling the impression-click-conversion pathways and learning across all exposure space via multi-task structure and transfer of representation, showing significant returns on the large-scale recommender/advertising data [13]. This contribution encourages a more general finding in the literature: conversion prediction cannot be viewed as a post-hoc modeling problem, but instead as an end-to-end funnel estimation problem, especially when the system goal is ranking, bidding or budget allocation based on sparse outcomes.

Lastly, operational reliability is pegged on quality ranking and also on quality. The modern neural models may not be well calibrated, which is problematic when predicted probabilities act as determinants of bidding multiplier, pacing determiners and ROI/CPA limits. Calibration experiments indicate that deep networks can be miscalibrated in making confidence estimates and that post-hoc lightweight methods (e.g., temperature scaling) can significantly enhance the reliability of probabilities [15]. This can be used in ad decision systems to justify the reviewer view that calibration is a first-class sense of evaluation, along with AUC/log-loss, in particular when models are being used as subsystems within constrained optimizers.

B. Incrementality, Causal Uplift and the Limits of Attribution-Optimized Learning.

The main point that has been made in the marketing analytics literature is that high forecasted conversion propensity does not correlate with high incremental impact. Attribution-based labels (e.g., last-click) artificially complicate the difference between exposure and user intent and selection, which provides models with incentives to find converters instead of persuadables. This encourages uplift modeling and causal inference as alternatives to supervised prediction, especially in a cost-effective manner in terms of budget and iROAS-like goals. Whereas supervised CVR models achieve predictive fit on observed results, causal/uplift procedures attempt to estimate heterogeneous treatment effects, the degree to which conversion probability varies in response to exposure, thereby and thus modeling matches the management aim of incremental value.

According to evidence given by deployed advertising ML, problem formulation and distribution shift are as important as algorithm choice. As an illustration, massive targeted display advertisement systems have used transfer learning across campaigns and goals, noting that the training allocation is frequently not equal to the decision environment, and the operational worth needs to be strongly formulated [14]. In this light of incrementality, this supports a major gap: many studies in the conversion prediction literature have found improvements when observing under an observational label, yet fewer studies have found that these improvements are also transferrable to incremental lift without randomized or quasi-experimental support.

As a reviewer, there is an indication in the literature that there are two viable implications of conversion optimization research. To begin with, research must differentiate between predictive performance and decision performance: an uplift-conscious system can not be a worse predictor on AUC but may perform better on incremental conversions per dollar. Second, plausible incrementality assertions usually need either controlled experimentation or high identification tactics. In turn, synthesized prediction, uplift estimation, and analysis methodology is worth reviewing due to the fact that it explicitly states the circumstances in which better prediction should and should not produce better marketing results.

C. Adaptive Decision Policies: Bandits, Constraints and Delayed Conversion Feedback.

Advertising optimization is both sequential and resource-constrained even in cases where incrementality is being estimated. There are spend budgets, pacing needs, CPA goals in campaigns, and the environment is not stationary because of the dynamics of an auction, creative exhaustion, and seasonal demand. This encourages the contextual formulations of bandits, where the system has to compromise exploration and exploitation without violating constraints. The bandits with knapsacks framework formalizes learning in the presence of a budget/resource constraint and it offers regret guarantees to the policies that are forced to allocate scarce resources to actions over time [16]. This kind of formulation applies more directly to ad allocation problems where the action in question is spend-consuming and the associated reward is a stochastic (and in many cases, delayed) reward.

Delayed conversions are a notably significant complication of operation: rewards are delayed and can bias learning, delay adaptation and cause distortion of offline assessment. The literature of online learning offers a systematic study of the effect of delay in increasing regret and also offers generic modifications to non-delayed algorithms to convert them into delayed-feedback ones [17]. In the case of conversion systems, it means that conversion lags, whether long or variable, can cause naive conversion of standard bandit updates applied to click-time signals to be unstable or biased. Thus, to lessen the risk of deployment, more recent ad decision architectures incorporate delayed-feedback processing, conservative exploration, and powerful off-policy assessment.

All in all, the analyzed literature leads to one and the same conclusion conversion prediction is a condition, but it is not enough to optimize advertising. The best direction of research is to integrate (i) structured supervised models to predict sparse conversion, (ii) incrementality measurement by causal/uplift methods, and (iii) adaptive and constrained decision policies, which explicitly consider delayed feedback. Evidence is also most fragmented in this integration, as methods are usually studied individually, which presents a definite incentive to review synthesis and standardized evaluation advice.

III. Background and Problem Formulation

A. Conversion Events, Prediction Targets, and Optimization Objectives

A conversion in digital advertising and mar-tech systems is a downstream user action of business value (i.e. purchase, registration, subscription, app install). The essence of the modeling is to estimate the conversion probability as given the user, context, and ad exposure signals. Practically, conversion prediction is seldom an objective in its own right: predictive probabilities are fed into decision layers (ranking, bidding, budget allocation, pacing) to maximize goal which can be expected value or CPA efficiency or incremental return. The most important complication is that conversion labels are generally gathered by observational logging and attribution regulations, which are not necessarily consistent with the causal aim of enhancing incremental conversions. This encourages a cautious problem formulation which isolates (i) estimating probability under observed exposure and (ii) estimating incremental impact and (iii) resource-constrained decision policies that are sequential in the face of uncertainty.

The second basic factor is the time-based aspect of conversion processes. Conversions are the result of a user journey which can involve both exposures and delays; hereby any given conversion probability is implicitly defined within a time window and protocol of observation. As a result conversion prediction in production should be considered more of a funnel and time to event estimation problem than a fixed time binary classification problem especially when the model outputs are directly influencing the real time allocation decision and must be robust to drift.

B. Data Challenges in Industry: Sparsity, Sample-Selection Bias and Delayed Feedback.

Three common challenges that define model choice and evaluation in conversion systems as reported in the literature include sparsity in data, sample-selection bias and delayed feedback. In most applications, impressions are rich, clicks are less, and conversions are meager; therefore, labels are few and there is extreme imbalance in classes. Sample-selection bias occurs due to the fact that standard CVR pipelines often use a biased sample of the traffic (e.g. clicked samples) to train but apply inference to the full impression space to give a distribution mismatch and biased parameter estimates. Late feedback further corruptions labels in training: often, conversions may take a stochastic delay to arrive, so that no conversion observed yet does not imply true negative: in particular, truncation of training data by fixed logging windows makes the labels corrupted.

Delayed feedback A traditional and powerful model of delayed feedback is the delayed feedback model of Chapelle, which is an explicit time-to-conversion model and conversion observational behavior is the function of conversion propensity and delay distribution [18]. The most important methodological implication is that no conversion probability and delay modeling should be done separately: the ignorance of delay systematically lowers eventual conversion probability and generates false-negative labels to samples that convert outside of the observation window [18]. More recent research confirms this point by updating that delayed feedback is interacting with selection bias and sparsity in real-world CVR. As an example, ESDF predicts the delay time based on the entitlement of an entire space and predicts delay times based on a framing of survival analysis to deal with delayed labels and also reduce sparsity and selection bias by designing the architecture and sharing of parameters [19]. Such findings substantiate a reviewer interpretation that much more frequently than a reviewer interpretation of just increasing model capacity are those that encode the actual data-generating constraints funnel structure and delay.

The combined occurrence of delayed feedback and selection bias also finds focus in more recent empirical studies involving combination of propensity-style corrections with likelihood-based delayed feedback modeling. Hu et al. specifically claim that methods that make use of reweighting to deal with exposure bias may still be biased when delayed conversions are not included, and suggest joint learning methods that make use of delayed-feedback likelihoods along with debiasing mechanisms [20]. These strands combined are an indication that the quality of prediction of conversion is to be measured not only by displaying discrimination measures but also by whether the training goal and labeling scheme are a good representation (i) of exposure mechanisms, (ii) of incomplete observation caused by delay, and (iii) of the target decision context.

C. Constraints of Measurement and Signal Loss due to Privacy.

Another quickly growing pressure is that conversion measurement is becoming more constrained by privacy constraints, further limiting observability at the user level, and restricting the ability to use cross-site identifiers. The Privacy Sandbox Attribution Reporting API in browser-based settings is being framed as a privacy-preserving feature to measure interactions between ads that result in conversions using browser-generated reports, and specifically seek to limit the use of invasive tracking while maintaining the ability to use attribution and measurement applications [21]. Modeling perspective This can be extended by such mechanisms modifying the granularity, timeliness and completeness of training signals, which can amplify label noise, delay effects, and aggregation bias. Conversion modeling studies should therefore expect that historical rich log assumptions do not necessarily work equally across platform and geography.

The privacy limitations also impel the learning paradigms that lower the amount of raw user information that is gathered centrally. Federated learning makes official a scenario where the update of models are learned locally on machines and aggregated centrally such that learning may use sensitive data without necessarily uploading it to a server [22]. Although the challenge of incrementality and attribution is not directly addressed by federated learning, it is also becoming increasingly applicable as a systems-level alternative when conversion signals are either fragmented or privacy-relevant or partially unavailable. Privacy-aware conversion optimization is better understood as a triad in terms of review; (i) limited measurement interfaces (e.g. aggregated reporting), (ii) effective modeling techniques that deal with sparsity and latency, and (iii) deployment systems that limit the needless exposure of raw behavioral data.

In general, the background literature points to an extensive synthesis conversion prediction is a biased-observation, delayed-label, sequential decision problem that is increasingly restricted on the measurement. Such techniques that explicitly model the funnel shape, manage latent feedback, rectify the selection mechanism, and are resistant to privacy-induced signal corruption tend to be more likely to be transformed into a practical indication of high-quality decision making [18]–[22].

IV. Machine Learning Approaches for Conversion Prediction

A. Classical Models

Generalized linear models (GLMs), usually logistic regression, are widely used as the starting point of classical conversion prediction pipelines because they are scalable and interpretable and their probability output is well-behaved. Thin one-hot encodings and constructed cross-features in recent high-dimensional advertising logs have long been used in the role of enabling GLMs to be competitive, especially where the operational requirements are biased towards stable coefficients and explainability. Nonetheless, they have a restricted capacity to model the interaction of complex features (user–ad-context) without a great deal of manual feature engineering due to their linear decision surface. With the growth of conversion systems to larger and more heterogeneous inventories, the literature started to think more and more of classical models as strong baselines that were however inadequate to describe the non-linear interaction structure at scale, leading to ensembles and neural architectures in the discovery of representation learning and interaction discovery [10], [11], [13].

B. Tree-Based Ensembles

Gradient-boosted decision trees (GBDTs) and tree-based ensembles are popularly used to perform conversion modeling on ad-tech (tabular) data due to their inherent ability to capture non-linearities, support mixed types of features and their good performance even with moderate-level feature engineering. XGBoost proposed a scaled, sparsity-conscious boosting system that has engineering-optimizations that enabled boosted trees to be practical in large-scale production environments and has continued to be a baseline of high-performance tabular prediction [23]. LightGBM also focused on high-dimensional and large data with histogram-based learning and leaf-wise growth plans, to ensure large speedups in training, and still achieve competitive accuracy, which is useful when retraining is required in non-stationary advertising settings [24].

In reviewer terms, it is unanimously empirically observed that GBDT variants tend to achieve a high classical ML ceiling on structured logs, particularly with feature interactions being significant, and with the system not having sufficient training data to utilize deep models. Their weaknesses are also very obvious: they can be hard or even impossible to scale to very high-cardinality categorical features without careful design and they do not naturally model long sequences of behavior, which may be valuable in journeys-aware conversion modeling [12], [13].

C. Deep Learning Methods

The conversion prediction methods that use deep learning are mostly driven by the necessity to learn non-numerical categorical variable embeddings and to capture the high-order interactions, without cross-enumeration. DeepFM is an example architecture that jointly applies factorization-machine type low-order interactions with deep networks to higher-order interactions with excellent CTR/CVR results on sparse feature spaces [10]. DCN offers a precise cross network which establishes feature interaction terms of limited degree, and offers an efficient trade off between the expressiveness and controllable complexity in production ranking systems [11].

In addition to the interactions between fixed features, behavior-sensitive deep models underline that the propensity to conversion relies on the history of actions of a user and their belonging to the candidate ad. DIN identifies this by acquiring an adaptive interest representation as conditioned on the current target according to attention-like processes across a history of user behavior, and it is better able to predict heterogeneous user intent in large-scale commercial contexts [12]. Moreover, deep models are also being developed to tackle the funnel structure and selection bias, redefining CVR as an end-to-end sequence model (impression → click → conversion), and ESMM shows that multi-task model can eliminate bias in training on clicked-only samples and can also more effectively predict post-click conversion on the overall impression space [13].

However, deep models present different operational risks: they are susceptible to change of distribution, need to be regularly and closely regularised and monitored, and importantly, they generate miscalibrated probability estimates at a time when ranking metrics might be improving. The use of calibration is thus not optional when the predicted conversion probabilities are spent on bidding, pacing and ROI/CPA decision layers [15].

D. Class Imbalance and Calibration.

The prediction of conversion is structurally asymmetrical: conversion is low, compared to impressions and frequently low, compared to clicks. Imbalance has an impact on dynamics of learning and evaluation. The most common mitigation methods are class-weighted losses, negative downsampling with cautious correction and focus on hard-examples. A canonical formulation known as focal loss up-weights informative examples, and down-weights easy negatives, and the principle underlying it applies to extreme imbalance regimes of conversion modeling (even in task-specific variants) directly [25]. Meanwhile, due to delayed conversions, temporary negatives may be introduced in fixed observation windows, and label quality is further complicated, delayed-feedback modeling specifically considers conversion observation as time-to-event instead of a final discrete outcome [18], [19].

Calibration is also central since the decision systems respond to the probabilities and not just ranks. Recent advances in modern neural networks provide evidence that large capacity models are not well calibrated, leading to the proposal of post-hoc correction (e.g. temperature scaling) and calibration-aware validation strategies to guarantee that model predictions approximate empirical frequencies [15]. When applied to conversion systems, it can be translated to a practical recommendation: consider discrimination (AUC/PR-AUC/log-loss) and probability reliability (calibration error, reliability curves), and consider monitoring calibration drift as a part of deployment readiness, especially when upon CPA/ROAS targets, the outputs are used in bid shading, pacing controllers, or constraint satisfaction.

Comparison of Machine Learning Approaches for Conversion Prediction

Model Category	**Example Models**	**Strengths**	**Limitations**	**Best Use Case**
Classical Models	Logistic Regression	Simple, interpretable, fast	Cannot capture complex interactions	Baseline models, small datasets
Tree-Based Models	XGBoost, LightGBM	Strong performance on tabular data, handles non-linearity	Limited sequence modeling	Large-scale ad data
Deep Learning Models	DeepFM, DCN, DIN	Captures high-order interactions, learns embeddings	Requires large data, less interpretable	Complex user behavior modeling
Funnel-Based Models	ESMM	Reduces bias, models full user journey	More complex architecture	Click → conversion pipelines
Delay-Aware Models	Delayed feedback models	Handles late conversions	Increased model complexity	Real-time ad systems

V. CAUSAL UPLIFT MODELING AND INCREMENTALITY

A. Attribution vs. Incrementality.
The majority of conversion pipelines maximize the outcome being attributed, but the attribution gives the credit without determining causation. This allows models that maximize attributed conversions to allocate budget to users with high baseline propensity in a systematic way instead of users in which exposure increases its conversion probability in a mode that makes sense. This creates ineffectiveness whereby the target of operation is incremental conversions or iROAS. Uplift modeling tackles this goal discongruity, estimating heterogeneous causal effects, or the change in outcome due to treatment, to address the goal of estimating the model in line with causal KPIs and making decisions with causal proxies [4], [26].

B. Frameworks and Meta-Learners of Uplift Estimation.
Uplift modeling is commonly formalized via the Conditional Average Treatment Effect **(CATE)**, $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$. One practical and common way to estimate CATE could be the meta-learner paradigm, which causes the estimation of the causal effect to be supervised learning. Künzel et al. make a unitary treatment of S-, T-, and X-learners and demonstrate that the X-learner is beneficial when there is treatment-control imbalance, and response functions have exploitable structure [26]. Complementary, causal forests are extensions of tree ensembles to CATE estimation with principled splitting and inference-based design, which is a scalable model of heterogeneity of effects with control over overfitting using honesty and sample splitting [27]. These techniques, together, make it very clear that uplift performance is neither simply a matter of model class, nor a matter of regularization, but a matter of the quality of the estimation of the nuisance, or of the validity of the assumptions under which the data was actually generated [4], [26], [27].

C. Bias Correction and Robust Estimation in the Non-Random Exposure.
Whether in the context of advertising data exposure, bidding, targeting and platform delivery mechanisms often confound exposure, naive uplift estimates are thus biased in an observational context. Strong methods usually use a combination of outcome and propensity estimation to make them less sensitive to the nuisance error. R-learner articulated by Nie and Wager, builds a residual-on-residual objective that isolates signal of treatment-effort and may have an improved statistical property when the nuisance models are sufficiently accurate [28]. In case of conversion systems, label incompleteness due to delayed conversion should also be considered as robustness: in the event that results are truncated by observation time, propensity models and outcome models can be biased, and this bias is transferred to CATE outcomes. Delayed-feedback modeling offers a clear structure of treating conversion observation as a time-to-event, as opposed to a final binary label, operationally applicable when uplift scores are being calculated and utilized prior to outcomes being fully matured [18], [19]. The methodological implication is that to estimate credibility in production, joint consideration of confounding, censoring/delay and drift is necessary; independent consideration of these variables in production is usually insufficient.

D. Uplift and Incrementality Evaluation.
Since counterfactuals cannot be observed at the individual level, uplift evaluation will have to be based on randomized designs or quasi-experimental estimators and ranking-based criteria. Qini -type curves, and related summary statistics evaluate the ranking of units within a model in a way that incremental utility is concentrated in the highest-ranked strata, which is a decision-relevant representation of model utility [29]. Recent statistical methods improve Qini-based regression and elucidate estimation and comparison in the presence of uplift goals [30]. Offline uplift metrics are not however a substitute of causal validation: the most robust evidence is now incrementality experiments (holdouts, geo experiment or randomized delivery), which directly estimate lift under realistic conditions. In this regard, uplift reporting is supposed to bridge (i) incremental returns in operational targeting depths, (ii) threshold and budget limits sensitivity and (iii) non-stationary stability under delayed feedback. Regarding reviewer terminology, an uplift model can be convincing to the extent that it is more effective in improving incremental results per unit of expenditure in plausible identification and deployment-relevant operating environments.

Comparison of Attribution-Based Prediction and Causal Uplift Approaches

Approach	**Target Objective**	**Key Advantage**	**Limitation**	**Business Relevance**
Attribution Models	Predict observed conversions	Easy to implement	Ignores causality	Limited optimization
Predictive ML Models	Conversion probability	High accuracy	May target “sure converters”	Moderate
Uplift Models	Incremental conversions	Captures true ad impact	Needs treatment/control data	High
Meta-Learners	CATE estimation	Flexible, scalable	Sensitive to bias	High
Causal Forests	Heterogeneous effects	Handles complex patterns	Computational cost	High

VI. CONTEXTUAL BANDITS FOR DYNAMIC AD ALLOCATION

A. Motivation Scoring to Sequential Control.
The conversion prediction provides a score; an advertising delivery needs a policy. In the operating environment, decisions (bids, placements, creatives, audience segments) have to be chosen over and over again in the face of

uncertainty and the environment changes and the reward signal is sparse. Contextual bandits formalize it as sequential covariated decision-making, in which the task is to maximise overall reward by exploitation of high-performing actions and exploration to learn and adapt [6]. This framing is especially adequate in cases where complete RL state modelling is not needed, and myopic greedy optimization can be unreliable or biased.

B. Core Policies and Offline Evaluation.

The literature provides two principles of the reviewer-critical conditions of bandit use in advertising (i) principled exploration and (ii) plausible offline policy evaluation (OPE) before use. Replay-based evaluation offers a quantitative, honest approach to the comparison of policies using recorded interaction information that eliminates the need to use simulators that create modeling bias [6]. In addition to replay, doubly robust estimators minimize the variancebias trade-off of OPE by jointly estimating reward models as well as propensity correction and they are consistent when either reward modeling or logging policies is specified imperfectly, which is desirable in ad systems when both are imperfect [5]. A publishable bandit study in this area ought to then report not just online results, but also logging assumptions, propensities and OPE diagnostics.

C. Budget/CPA/Pacing: as Constrained Bandits.

Practical ad placement is limited to the use of spend budgets, pacing schedules and CPA/ROAS targets. The Bandits with knapsacks offers a canonical model of contextual decision making when resources are scarce, and the resulting policies are explicit in terms of consumption of limited budgets and learning action values [16]. Such constraint-based framing is more justifiable than post hoc intuitions, since constraint-based framing puts feasibility into the learning goal and makes the theoretical definition of optimal when there are budget constraints clear. To be accepted by the reviewer, the mapping between business constraints (budget, CPA) on the one hand and resource consumption and reward on the other hand should be expressed explicitly, as well as the way constraints are imposed (hard/soft) and the way feasibility violations are managed.

D. Delays conversion and Censoring.

The reward of the advertising is often delayed: the clicks will be instant, conversions can take place hours or days later, and delayed events can be blocked by attribution windows. Late feedback has a significant impact on the dynamics of learning and may cause systematic bias when the message of not yet observed is perceived as negative. It has been analyzed that online learning experiences delay which causes regret and needs tuning algorithmically instead of some slight adjustments [17]. Modeling conversion delays conversion specific bandit models use conversion delays to model conversions in idealized fully observed environments, contrasting with realistic censoring, and offer low conversion rate tailored algorithms to delayed observation [7]. A stringent treatment must hence be a concrete model to delay/censoring and metrics sensitivity to attribution windowing as well as maturation period.

VII. Evaluation, Privacy, and Deployment Considerations

The study of conversion optimization will have to isolate predictive quality, causal decision quality, and operational reliability. In predictive models, the metrics of discrimination like AUC or PR-AUC can be informative although they are not complete when there is extreme imbalance between classes; PR-AUC is often more informative when positives are rare [40]. The bidding and constraint controllers in production consume probability outputs, and thus, calibration is necessary. Even improved ranking can systematically be miscalibrated with an improved modern deep model; instead, calibration-aware validation and post-hoc correction must be considered deployment requirements instead of optional reporting [15]. Moreover, the fact that feedback is delayed leads to label censoring in a fixed observation window; delayed-feedback and survival-style formulations are needed to prevent training target and downstream evaluation biasing [18], [19], [42].

User / Context Input
↓
Feature Extraction
↓
Prediction Model
(Conversion / Uplift Score)
↓
Decision Layer
(Contextual Bandit / Policy)
↓
Constraint Layer
(Budget / CPA / Pacing)
↓
Ad Allocation
↓
User Interaction
↓
Delayed Conversion Feedback
↓
Model Update (Learning Loop)

Integrated Conversion Prediction and Decision-Making Framework

In the case of causal and bandit elements, the assessment should not be limited to supervised measures. There is offline policy evaluation (OPE). It is an important protection in the event of limited experimentation budget or risk tolerance. Replay-based assessment gives an objective framework in the correct logging conditions, and doubly robust estimators decrease Auscultation and stay consistent in case either the reward model or propensity model is entirely detailed [6], [5]. In reasonable cases, randomized lift tests are the most informative evidence on incrementality; observational adjustment using propensity methods is informative, but relies on assumptions that cannot be tested and must be discussed in terms of sensitivity [43].

The limitations on privacy were increasingly placing a limit on what is measurable and learnable. Brower-level and platform-level modifications have shifted the measurement to privacy-sensitive reporting APIs (e.g., Attribution Reporting API) and out of unlimited user-level tracking [9], [48]. Limitations in mobile measurement equally encourage the use of privacy-controlling, aggregate schemes like

SKAdNetwork [49] and explicit consent regimes like App Tracking Transparency [8] instead. With these conditions, the design of the model should expect weaker, delayed and aggregated labels, which leads to the importance of robust delayed-feedback modeling and conservative decision policy.

Lastly, drift and failure mode should be explicitly monitored in the deployment reliability category. Non-stationary of auction market and users behavior requires a regular recalibration and retraining, and strict controls on feature soulage, maturity windows and stability limits. As a standard of review, a work that purports to have business impact must describe the entire measurement protocol (windows, attribution rules, and censoring handling), the OPE/experiment design, which will be used to validate it, and the stability controls that will eliminate regressions on performance after launch [15], [18], [6].

VIII. Research Gaps and Future Directions

Existing studies show that there is a lot of research done in terms of conversion prediction, uplift estimation, constrained ad allocation; but, the literature is still disjointed in terms of these elements. The first significant area of gap is persistence of the disjunct between predictive accuracy and incremental decision quality. Numerous studies continue to find improvements in AUC, log-loss, or ranking performance without determining whether the improvement is associated with increased causal lift, reduced acquisition cost, or increased iROAS when deployed in practice [1], [4], [13], [25], [45]. Consequently, there remains a relative lack of unified framework in the field that brings the supervised conversion modeling, heterogeneous treatment-effect estimation and online allocation policy together in the same evaluative pipeline [4], [16], [26], [30].

The second gap is related to delayed and censored results. Even though delayed-feedback modeling has been considered in both conversion prediction and online learning, the largest practical systems continue to use truncated observation windows, which erroneously label late conversions and bias predictive and causal estimates [7], [17]–[20]. The work done in the future should hence emphasize on doing the modeling of delay, censoring, and treatment effect heterogeneity together instead of tackling them separately [18]–[20], [27], [28].

The third issue not yet solved is robustness in the case of measurement that is subject to privacy constraints. Well beyond browser-level and platform-level constraints, user-level attribution faces more and more constraints, and signal granularity is diminished, but much of the literature still presupposes rich, persistent tracking data [8], [9], [21], [48], [49]. This causes an urgent requirement of conversion systems that can be trusted in aggregated reporting, sparse feedback, and biased observability, possibly with federated learning, privacy-preserving causal estimation, and calibration-aware decision layers [15], [21], [22].

Lastly, future studies must put more stress on deployment-valid assessment. They are useful, e.g. offline policy evaluation, uplift ranking metrics and benchmark prediction scores, but not replacements of stable online performance under budget, pacing and CPA constraints [5], [6], [16], [28], [29]. In the view of the reviewer, the most publishable next-generation work will be one incorporating prediction, incrementality, delayed feedback and constrained exploration into a coherent, privacy conscious decision architecture with both rigorous offline diagnostics and plausible online validation [5] - [7], [16], [17], [30].

IX. Conclusion

Conversion prediction has developed beyond a limited post-click classification challenge to a more general challenge of causal, sequential and privacy constrained decision-making. In the literature examined in this paper, it is demonstrated that classical and tree-based models continue to be useful in structured advertising data due to their efficiency and stability in operations whereas deep learning methods are better in terms of representation learning and interaction modeling in sparse and high-dimensional contexts. Meanwhile, predictive accuracy is not enough in the case of the managerial goal, which is incremental value and not attributed volume. It is the reason why uplift modeling, heterogeneous treatment-effect estimation and robust causal evaluation have taken center stage in conversion optimization studies nowadays.

The review also suggests that systems of deployment quality should be able to interact prediction with adaptive allocation schemes like contextual bandits, in particular when there are budget, pacing, and CPA limits and when the conversion feedback is delayed. Another characteristic difficulty is the granular-measurement loss in privacy-preserving advertising systems that demands models and evaluation pipes that are consistent in conditions of partial observability and limited attribution indicators.

In general, no individual, better algorithm is the most promising, but rather a combined framework, integrating credible incrementality estimation, calibrated prediction, constrained sequential optimization, and privacy-aware measurement. Future practice ought to thus give priority to deployable architectures which combine causality, delayed feedback and signal degradation in real world advertisement systems.

References

[1] T. Xue, Y. Yang, and P. Zhai, "Conversion rate prediction in online advertising: modeling techniques, performance evaluation and future directions," *arXiv preprint* arXiv:2512.01171, 2025. [Online]. Available: https://arxiv.org/abs/2512.01171

[2] H.-T. Cheng *et al*., "Wide & Deep Learning for Recommender Systems," *arXiv preprint* arXiv:1606.07792, 2016. [Online]. Available: https://arxiv.org/abs/1606.07792

[3] O. Chapelle, E. Manavoglu, and R. Rosales, "Simple and Scalable Response Prediction for Display Advertising," *ACM Trans. Intell. Syst. Technol.*, vol. 5, no. 4, 2014, doi: 10.1145/2532128. [Online]. Available: https://dl.acm.org/doi/10.1145/2532128

[4] P. Gutierrez and J.-Y. Gérardy, "Causal Inference and Uplift Modelling: A Review of the Literature," in *Proc. PMLR*, vol. 67, pp. 1–13, 2017. [Online]. Available: https://proceedings.mlr.press/v67/gutierrez17a.html

[5] M. Dudík, J. Langford, and L. Li, "Doubly Robust Policy Evaluation and Learning," *arXiv preprint* arXiv:1103.4601, 2011. [Online]. Available: https://arxiv.org/abs/1103.4601

[6] L. Li, W. Chu, J. Langford, and X. Wang, "Unbiased Offline Evaluation of Contextual-bandit-based News Article Recommendation

Algorithms," *arXiv preprint* arXiv:1003.5956, 2010. [Online]. Available: https://arxiv.org/abs/1003.5956

[7] C. Vernade, O. Cappé, and V. Perchet, "Stochastic Bandit Models for Delayed Conversions," *arXiv preprint* arXiv:1706.09186, 2017. [Online]. Available: https://arxiv.org/abs/1706.09186

[8] Apple, "App Tracking Transparency | Apple Developer Documentation," 2026. [Online]. Available: https://developer.apple.com/documentation/apptrackingtransparency

[9] Google, "Overview of Attribution Reporting API | Privacy Sandbox," updated Dec. 18, 2025. [Online]. Available: https://privacysandbox.google.com/private-advertising/attribution-reporting

[10] H. Guo, R. Tang, Y. Ye, Z. Li, and X. He, "DeepFM: A Factorization-Machine based Neural Network for CTR Prediction," *arXiv preprint* arXiv:1703.04247, 2017. [Online]. Available: https://arxiv.org/abs/1703.04247

[11] R. Wang, B. Fu, G. Fu, and M. Wang, "Deep & Cross Network for Ad Click Predictions," in *Proc. ADKDD (KDD Workshop)*, 2017, doi: 10.1145/3124749.3124754. [Online]. Available: https://dl.acm.org/doi/10.1145/3124749.3124754

[12] G. Zhou *et al.*, "Deep Interest Network for Click-Through Rate Prediction," in *Proc. ACM SIGKDD*, 2018, doi: 10.1145/3219819.3219823. [Online]. Available: https://dl.acm.org/doi/10.1145/3219819.3219823

[13] X. Ma *et al.*, "Entire Space Multi-Task Model: An Effective Approach for Estimating Post-Click Conversion Rate," in *Proc. ACM SIGIR*, 2018, doi: 10.1145/3209978.3210104. [Online]. Available: https://dl.acm.org/doi/10.1145/3209978.3210104

[14] C. Perlich, B. Dalessandro, T. Raeder, O. Stitelman, and F. Provost, "Machine learning for targeted display advertising: Transfer learning in action," *Machine Learning*, vol. 95, no. 1, pp. 103–127, 2014, doi: 10.1007/s10994-013-5375-2. [Online]. Available: https://link.springer.com/article/10.1007/s10994-013-5375-2

[15] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On Calibration of Modern Neural Networks," *arXiv preprint* arXiv:1706.04599, 2017. [Online]. Available: https://arxiv.org/abs/1706.04599

[16] S. Agrawal, N. R. Devanur, and L. Li, "An efficient algorithm for contextual bandits with knapsacks, and an extension to concave objectives," in *Proc. COLT*, *Proc. PMLR*, vol. 49, pp. 4–18, 2016. [Online]. Available: https://proceedings.mlr.press/v49/agrawal16.html

[17] P. Joulani, A. György, and C. Szepesvári, "Online Learning under Delayed Feedback," *arXiv preprint* arXiv:1306.0686, 2013. [Online]. Available: https://arxiv.org/abs/1306.0686

[18] O. Chapelle, "Modeling Delayed Feedback in Display Advertising," in *Proc. ACM SIGKDD*, 2014, doi: 10.1145/2623330.2623634. [Online]. Available: https://dl.acm.org/doi/10.1145/2623330.2623634

[19] Wang, Z. Wang, X. Zhu, and K. Gai, "Delayed Feedback Modeling for the Entire Space Conversion Rate Prediction," *arXiv preprint* arXiv:2011.11826, 2020, doi: 10.48550/arXiv.2011.11826. [Online]. Available: https://arxiv.org/abs/2011.11826

[20] T. Hu *et al.*, "Debiasing the Conversion Rate Prediction Model in the Presence of Delayed Implicit Feedback," 2024. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC11431287/

[21] H. B. McMahan *et al.*, "Communication-Efficient Learning of Deep Networks from Decentralized Data," in *Proc. AISTATS*, *Proc. PMLR*, vol. 54, pp. 1273–1282, 2017. [Online]. Available: https://proceedings.mlr.press/v54/mcmahan17a.html

[22] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proc. ACM SIGKDD*, 2016, doi: 10.1145/2939672.2939785. [Online]. Available: https://dl.acm.org/doi/10.1145/2939672.2939785

[23] G. Ke *et al.*, "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," in *Proc. NeurIPS*, 2017. [Online]. Available: https://proceedings.neurips.cc/paper/6907-lightgbm-a-highly-efficient-gradient-boosting-decision-tree.pdf

[24] T.-Y. Lin *et al.*, "Focal Loss for Dense Object Detection," in *Proc. IEEE ICCV*, 2017. [Online]. Available: https://arxiv.org/abs/1708.02002

[25] S. R. Künzel, J. S. Sekhon, P. J. Bickel, and B. Yu, "Metalearners for estimating heterogeneous treatment effects using machine learning," *PNAS*, vol. 116, no. 10, pp. 4156–4165, 2019, doi: 10.1073/pnas.1804597116. [Online]. Available: https://www.pnas.org/doi/10.1073/pnas.1804597116

[26] S. Wager and S. Athey, "Estimation and inference of heterogeneous treatment effects using random forests," *JASA*, vol. 113, no. 523, pp. 1228–1242, 2018, doi: 10.1080/01621459.2017.1319839. [Online]. Available: https://www.tandfonline.com/doi/abs/10.1080/01621459.2017.1319839

[27] X. Nie and S. Wager, "Quasi-oracle estimation of heterogeneous treatment effects," *arXiv preprint* arXiv:1712.04912, 2017. [Online]. Available: https://arxiv.org/abs/1712.04912

[28] P. D. Surry and N. J. Radcliffe, "Quality measures for uplift models," 2011. [Online]. Available: https://www.stochasticsolutions.com/pdf/kdd2011late.pdf

[29] M. Belbahri, A. Murua, A. Gandouet, and V. Partovi Nia, "Qini-based uplift regression," *Ann. Appl. Stat.*, vol. 15, no. 3, 2021, doi: 10.1214/21-AOAS1465. [Online]. Available: https://projecteuclid.org/journals/annals-of-applied-statistics/volume-15/issue-3/Qini-based-uplift-regression/10.1214/21-AOAS1465

[30] D. Russo, B. Van Roy, A. Kazerouni, I. Osband, and Z. Wen, "A tutorial on Thompson sampling," *Found. Trends Mach. Learn.*, vol. 11, no. 1, pp. 1–96, 2018, doi: 10.1561/2200000070. [Online]. Available: https://www.nowpublishers.com/article/Details/MAL-070

[31] T. Lattimore and C. Szepesvári, *Bandit Algorithms*. Cambridge Univ. Press, 2020. [Online]. Available: https://www.cambridge.org/core/books/bandit-algorithms/8E4FBD7F1E0D05E8E6C2E6A6A5C6A0A3

[32] Y. Abbasi-Yadkori, D. Pál, and C. Szepesvári, "Improved algorithms for linear stochastic bandits," in *Proc. NeurIPS*, 2011. [Online]. Available: https://proceedings.neurips.cc/paper/2011/hash/e1d5be1c7f2f456670de3d53c7b54f4a-Abstract.html

[33] S. Agrawal and N. Goyal, "Thompson sampling for contextual bandits with linear payoffs," in *Proc. ICML*, *Proc. PMLR*, vol. 28, 2013. [Online]. Available: https://proceedings.mlr.press/v28/agrawal13.html

[34] S. Agrawal and N. Goyal, "Analysis of Thompson sampling for the multi-armed bandit problem," in *Proc. COLT*, *Proc. PMLR*, vol. 23, 2012. [Online]. Available: https://proceedings.mlr.press/v23/agrawal12.html

[35] H. B. McMahan *et al.*, "Ad Click Prediction: a View from the Trenches," in *Proc. ACM SIGKDD*, 2013, doi: 10.1145/2487575.2488200. [Online]. Available: https://dl.acm.org/doi/10.1145/2487575.2488200

[36] Y. Juan, Y. Zhuang, W.-S. Chin, and C.-J. Lin, "Field-aware Factorization Machines for CTR Prediction," in *Proc. ACM RecSys*, 2016. [Online]. Available: https://www.csie.ntu.edu.tw/~cjlin/papers/ffm.pdf

[37] S. Rendle, "Factorization Machines," in *Proc. IEEE ICDM*, 2010, doi: 10.1109/ICDM.2010.127. [Online]. Available: https://ieeexplore.ieee.org/document/5694074

[38] J. H. Friedman, "Greedy Function Approximation: A Gradient Boosting Machine," *Ann. Stat.*, vol. 29, no. 5, pp. 1189–1232, 2001. [Online]. Available: https://projecteuclid.org/journals/annals-of-statistics/volume-29/issue-5/Greedy-function-approximation-A-gradient-boosting-machine/10.1214/aos/1013203451

[39] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, 2002. [Online]. Available: https://jair.org/index.php/jair/article/view/10302

[40] J. Davis and M. Goadrich, "The Relationship Between Precision-Recall and ROC Curves," in *Proc. ICML*, 2006. [Online]. Available: https://dl.acm.org/doi/10.1145/1143844.1143874

[41] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant, *Applied Logistic Regression*, 3rd ed. Wiley, 2013. [Online]. Available: https://onlinelibrary.wiley.com/doi/book/10.1002/9781118548387

[42] D. R. Cox, "Regression Models and Life-Tables," *J. R. Stat. Soc. B*, vol. 34, no. 2, pp. 187–220, 1972. [Online]. Available: https://www.jstor.org/stable/2985181

[43] P. R. Rosenbaum and D. B. Rubin, "The Central Role of the Propensity Score in Observational Studies for Causal Effects," *Biometrika*, vol. 70, no. 1, pp. 41–55, 1983. [Online]. Available: https://academic.oup.com/biomet/article/70/1/41/240879

[44] J. Pearl, *Causality: Models, Reasoning, and Inference*, 2nd ed. Cambridge Univ. Press, 2009. [Online]. Available: https://www.cambridge.org/core/books/causality/0B3A25C5A0B4D7D6F9D9B3E1D0E3B1E9

[45] S. Athey and G. W. Imbens, "Machine Learning Methods for Estimating Heterogeneous Causal Effects," *Statistical Science*, vol. 31, no. 3, pp. 412–431, 2016. [Online]. Available: https://projecteuclid.org/journals/statistical-science/volume-31/issue-3/Machine-Learning-Methods-for-Estimating-Heterogeneous-Causal-Effects/10.1214/16-STS571

[46] A. Auer, P. Cesa-Bianchi, and P. Fischer, "Finite-time Analysis of the Multiarmed Bandit Problem," *Machine Learning*, vol. 47, pp. 235–256, 2002. [Online]. Available: https://link.springer.com/article/10.1023/A:1013689704352

[47] E. Kaufmann, O. Cappé, and A. Garivier, "On the Complexity of Best-Arm Identification in Multi-Armed Bandit Models," *JMLR*, vol. 17, no. 1, pp. 1–42, 2016. [Online]. Available: https://jmlr.org/papers/v17/14-301.html

[48] Google, "Get started with attribution reporting," *Privacy Sandbox*, Dec. 18, 2025. [Online]. Available: https://privacysandbox.google.com/private-advertising/attribution-reporting/getting-started

[49] Apple, "SKAdNetwork | Apple Developer Documentation," 2026. [Online]. Available: https://developer.apple.com/documentation/storekit/skadnetwork

[50] Microsoft Research, "LightGBM: A Highly Efficient Gradient Boosting Decision Tree (Publication page)," 2017. [Online]. Available: https://www.microsoft.com/en-us/research/publication/lightgbm-a-highly-efficient-gradient-boosting-decision-tree/